Venkata Lakshmi Parimala Pasupuleti

AI Engineer · RAG Pipelines · Agentic Workflows · LangChain · LangGraph · Document Intelligence · Prompt Engineering · Azure OpenAI · Python · Enterprise APIs · Vector Databases · SQL

+1-703-261-5410 | venkatalakshmiparimala@gmail.com | linkedin.com/in/venkata-lakshmi-parimala-pasupuleti | github.com/venkatalakshmiparimala/AgentProbe | Virginia, USA — Open to Relocation

PROFESSIONAL SUMMARY

AI engineer with 8 years of enterprise backend experience and hands-on production delivery of **RAG pipelines, agentic workflows, and document intelligence systems**. Built **Leasel**: a production RAG-based document intelligence and agentic investigation platform — **8-agent LangGraph system** (Triage → Retrieval → Configuration → Historical → Reasoning → Validation → Escalation → HITL) achieving 93.6% precision and 0% unsupported-claim rate on 1,240 compliance documents. Built **AgentProbe**: an AI evaluation platform with **RAG evaluation, multi-provider LLM routing** (OpenAI API / Azure OpenAI compatible, Anthropic, Groq, Ollama), 15-endpoint FastAPI backend, CI regression gate, and pass@k harness. At Entrata: **enterprise backend services, REST APIs, and microservices** serving 100K+ MAU at 99.5% uptime; delivered **scalable, production-ready AI solutions** (Leasel, AgentProbe); **200+ SQL query optimizations** (30% performance gain) across PostgreSQL and MySQL; **RabbitMQ async processing**; AWS S3 data integration pipelines with 100% PHI compliance. M.S. Data Analytics Engineering, George Mason University (GPA 3.87).

WORK EXPERIENCE

George Mason University

Aug 2024 – May 2026

Research Associate / Graduate Teaching Assistant — Python · SQL · Data Pipelines · Statistical Modeling

- Built multi-source Python data integration pipelines (190+ countries, WHO + UK Road Safety datasets) achieving **$R^2 = 0.977$**; applied Isolation Forest for automated data integrity validation across 3 research pipelines. Delivered Python and database design workshops to **600+ students**.

Entrata — Enterprise PropTech SaaS

Sep 2018 – Aug 2024 · Remote

Senior Software Engineer — Python · REST APIs · Microservices · SQL · AWS · Data Integration

- **Enterprise REST API and microservices backend**: 20+ production modules (Python, PHP, PostgreSQL, JavaScript) serving 100K+ MAU at 99.5% uptime; **RabbitMQ async processing** with retry logic and idempotency guarantees; **Redis caching** layer for high-frequency data access; Docker microservices redesign (50% scalability improvement, 30% deployment time reduction); CI/CD automation (GitHub Actions) with pre/post validation gates.
- **SQL and data integration**: optimized 200+ queries (30% performance gain) across PostgreSQL and MySQL via execution plan analysis, indexing, partitioning, stored procedures, triggers, and audit tables; owned schema design for 20+ modules; built **Python automation pipelines** transforming federal HUD source rows into ~180K domain records (eligibility, compliance, financial calculations) — eliminating 80% of manual operational workload.
- **AWS data integration**: led PHI-compliant AWS S3 migration (Libsodium encryption, zero security incidents); **300+ code reviews; mentored 5+ engineers**; built internal debugging and observability tooling reducing debugging time 40%.

Entrata India (Db Xento Systems)

Jan 2017 – Sep 2018 · Pune

Associate / Trainee Software Engineer — Python · SQL · REST APIs · AWS · Data Integration

- Built Python and SQL backend modules for property management and compliance; designed SQL data integration pipelines processing 1.6M federal housing records; led AWS S3 cloud migration; contributed to API development and defect analysis under senior engineer guidance.

PROJECTS

Leasel: RAG-Based Document Intelligence & Agentic Investigation Platform

RAG · AGENTIC

Python · LangGraph · LangChain · FastAPI · ChromaDB (Vector DB) · TF-IDF RAG · SQLite · Docker · Streamlit — github.com/venkatalakshmiparimala/Lease_complaince_rag

- **RAG-based document intelligence**: section-aware chunking, TF-IDF + semantic retrieval (ChromaDB vector database), 100% citation correctness, missing/conflicting clause detection, and human-review gate on regulated decisions — validated across 1,240 compliance documents (5 US states + federal).
- **Agentic workflow (LangGraph)**: 8-agent StateGraph (Triage → Retrieval → Configuration → Historical Incident → Structured Data → Reasoning → Validation → HITL Escalation) producing ranked, evidence-backed root-cause hypotheses with conditional escalation when confidence is too low. Core design: **LLM reasons; deterministic logic or human makes every regulated call**.
- Results: **93.6% precision, 0% unsupported-claim rate**; pass@1 81.7%, pass@3 88.3% via pass@k harness with multi-trial sampling and LLM-as-judge scoring.

AgentProbe: AI Evaluation Platform — RAG Evaluation · Multi-Provider LLM Routing · CI/CD

EVAL · RAG

Python · FastAPI · LangGraph · ChromaDB · React · Docker · CI/CD (GitHub Actions) — github.com/venkatalakshmiparimala/AgentProbe

- **RAG evaluation and AI platform**: 15-endpoint FastAPI backend; **multi-provider LLM routing** (OpenAI API / Azure OpenAI compatible, Anthropic, Groq, Ollama); 5-metric eval engine (correctness, groundedness, hallucination rate, latency, token efficiency); **CI regression gate** blocking quality regressions before merge; React observability dashboard tracking per-run metrics, retrieval drift, and latency trends.
- Validated 1,240+ test cases: **pass@1 81.7%, pass@3 88.3%**; LLM-as-judge scoring with multi-trial sampling; zero-vendor-dependency design — runs fully locally at zero cost.

AI on FHIR: NLP-to-API Integration & Enterprise REST Backend

REST APIS · NLP

Python · Flask · REST APIs (FHIR) · Docker · spaCy · BERT · React — github.com/venkatalakshmiparimala/AI_on_FHIR

- Python Flask backend integrating with FHIR REST APIs — **prompt-engineered NLP pipeline** translating natural-language queries into structured API calls (BERT + spaCy + LLM); HIPAA-compliant RBAC; Docker containerized deployment; ~60% reduction in manual clinical retrieval time. Demonstrates **end-to-end enterprise API integration with AI-driven NLP layer**.

EDUCATION

M.S. Data Analytics Engineering — George Mason University — May 2026 · GPA 3.87

AI/ML systems, data engineering, statistical modeling, RAG pipelines, NLP, performance engineering

B.Tech Computer Science & Engineering — JNTU Anantapur — 2013 – 2017

TECHNICAL SKILLS

Engineering & Backend: Python (expert), SQL (expert — 200+ production optimizations, schema design, normalization, execution plan tuning, indexing, partitioning, stored procedures, triggers, audit tables), PL/SQL, PHP, JavaScript, TypeScript, R; FastAPI, REST APIs, Enterprise API Integration, Microservices, RabbitMQ async messaging, Redis (caching), Docker, CI/CD (GitHub Actions), AWS (S3, EC2, RDS, Glue), PostgreSQL, MySQL, Oracle, SQLite, Git, Linux, Agile/Scrum

AI & Agentic: RAG Pipelines, Vector Databases (ChromaDB), Agentic Workflows, LangChain, LangGraph, Multi-Agent Orchestration, Document Intelligence, Knowledge Retrieval, Semantic Search, Embeddings, Prompt Engineering (few-shot, chain-of-thought, output validation, context engineering), Azure AI Foundry & Azure OpenAI (API-compatible), LLM Integration (OpenAI · Anthropic · Groq · Ollama), HITL, Confidence Scoring, Guardrails, Responsible AI, React, Streamlit, NLP (spaCy, BERT), MCP

ML & Analytics: PyTorch, scikit-learn, XGBoost, Random Forest, Isolation Forest, Feature Engineering, Anomaly Detection, Statistical Modeling, Model Evaluation, Tableau, Power BI, ggplot2, Planning Poker; 300+ code reviews, 5+ engineers mentored